

Kartik Choudhary

[LinkedIn](#) | [Github](#) | [LeetCode](#) | [Portfolio](#)

TECHNICAL SKILLS

Languages: Python [Advanced], C++ [Intermediate], Java[Basic], based on Kalvium Dojo Belt Levels Level Indicators - DOJO - 1 to 3 Basic ; DOJO - 4 & 5 Intermediate ; DOJO - 6 Advanced

Frontend: HTML, CSS, JavaScript, React.js, Tailwind CSS

Backend: Node.js, Express.js, REST APIs, JWT Authentication, RBAC, Prisma ORM, Redis, BullMQ

Databases: PostgreSQL, MongoDB, MySQL, Redis

API Technologies: OpenAI API, Google Vertex AI, Prompt Engineering, Embeddings, Vector Search, Retrieval-Augmented Generation (RAG), OAuth, JSON

Tools: VS Code, Postman, Docker, Google Colab, Vertex AI Studio, Render, Vercel, Git, GitHub, Docker Compose

Libraries: Mongoose, Socket.IO, bcrypt, Multer, dotenv, CORS, NumPy, Pandas, Scikit-learn

Concepts: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, System Design, Authentication, Authorization, Caching, Asynchronous Processing, Machine Learning Lifecycle, AI Agent Workflows

PERSONAL STATEMENT

Full-Stack and AI Developer experienced in building scalable web applications using **TypeScript, Python, React.js, Node.js, Express.js, PostgreSQL, MongoDB, Redis, and Docker**. Built **5+ production-ready projects**, including **LinkForge**, with **42+ REST APIs** and **30%** backend performance improvement through caching and query optimization. Solved **210+ LeetCode problems** and developed AI-powered applications using **LLMs** and **RAG**. Seeking a **Software Development Engineer Intern** role.

PROJECTS

LinkForge (Smart Link Management SaaS) | [Github Repo](#) | [Deployment Link](#) **June'26 - Present**

What you did: Built a **production-ready Smart Link Management SaaS** with **5+ core modules** and **42+ REST APIs**, enabling secure authentication, intelligent URL redirection, and real-time analytics.

- **Tech Stack and Frameworks used:** React.js, TypeScript, Node.js, Express.js, PostgreSQL, Prisma ORM, Redis, BullMQ, JWT Authentication, TanStack Query, Tailwind CSS, Docker, Git, GitHub.
- **Problem Solved:** Developed a scalable platform for businesses and developers to create, manage, and track smart links with **10+ real-time analytics metrics**, reducing **URL redirect latency by 30%** and **backend processing time by 35%**.
- **Challenges Tackled:** Architected an **event-driven analytics pipeline** using **Redis** and **BullMQ**, implemented **JWT-based RBAC**, and optimized **PostgreSQL** queries to improve scalability and backend performance.

Learning Management System(LMS) | [Github Repo Link](#) | [Deployment Link](#) **Jan'26 - July'26**

What you did: Built a **production-ready AI-assisted Learning Management System** with **2 role-based portals** and **20+ REST APIs**, enabling secure authentication, course creation, student enrollment, and personalized learning.

- **Tech Stack and Frameworks used:** React.js, TypeScript, Node.js, Express.js, MongoDB, JWT Authentication, RBAC, Tailwind CSS, Vite, Axios, REST APIs, Docker, Git, GitHub, Postman.
- **Problem Solved:** Developed a unified platform for educators and students, **reducing API response time by 25%** and **improving course management efficiency by 30%** through optimized MongoDB queries and scalable REST API architecture.
- **Challenges Tackled:** Implemented **JWT-based Role-Based Access Control (RBAC)**, optimized MongoDB schema design, integrated AI-assisted learning features, and strengthened expertise in scalable backend architecture, state management, and API design.

EDUCATION ENROLLMENT

Kalvium's UG program in CS (Software Product Engineering)

2024-28

Campus: Jaipur, Rajasthan | Bachelor's enrollment: B.Tech, JECRC University

ACHIEVEMENTS & EXTRACURRICULARS

- **Hackathons & Competitions:**

- [AI Hackathon Participation Certificate\(Kalvium\)](#) : Built a prototype using **Generative AI APIs** in under 24 hours, demonstrating rapid prototyping skills.
- [CodeRush Coding Competition\(unstop\)](#) ->Achieved **2nd place** out of **1000+ participants** by solving complex algorithmic challenges under timed conditions.

- **Technical Contributions:**

- [Google AI Essentials](#) : Completed Google's AI Essentials certification covering **Generative AI, Prompt Engineering, Responsible AI, AI Productivity, and 15+ hands-on practical exercises**, building foundational skills in effective AI-assisted workflows.
- [Copado AI Fundamentals](#) : Completed certification covering **7+ AI-powered DevOps capabilities**, including **AI Agents, CI/CD automation, testing workflows, Salesforce integration, and Slack-based development collaboration**.
- [Google Cloud – Prompt Design in Vertex AI](#) : Earned certification in AI prompt engineering, improving **accuracy, context relevance, and output quality by up to 20%** in LLM-based responses.
- [GSSOC 2025 - Participant](#) : Participated in the open-source event, exploring **10+ backend and full-stack repositories** to understand collaboration workflows, issue tracking, and version control.
- [Introduction to Generative AI Certificate](#) : **Completed** certification covering **LLM fundamentals, Prompt Engineering, AI APIs, and Generative AI concepts**, building **2+ AI-powered prototype applications**.
- [Google Cloud - Introduction to Generative AI](#) : Completed Google Cloud's introductory certification covering **LLM fundamentals, transformer architecture, generative AI concepts, and real-world enterprise applications**.